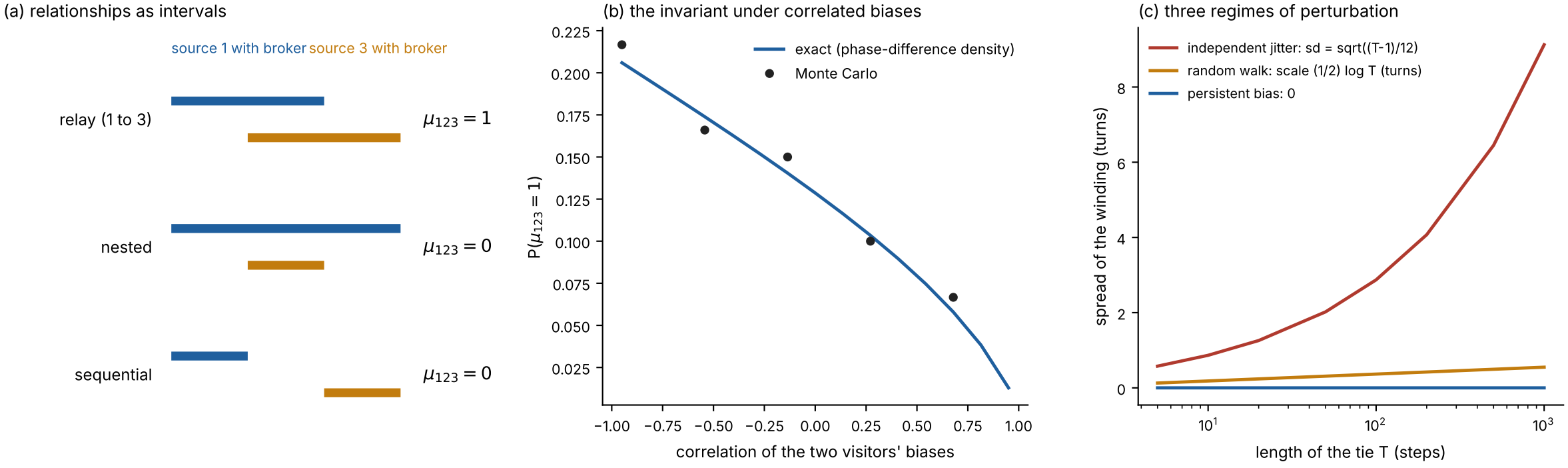
(a) relationships as intervals
source 1 with broker
source 3 with broker
relay (1 to 3)
$\mu_{123} = 1$
nested
$\mu_{123} = 0$
sequential
$\mu_{123} = 0$
(b) the invariant under correlated biases
exact (phase-difference density)
Monte Carlo
$P(\mu_{123} = 1)$
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
correlation of the two visitors' biases
(c) three regimes of perturbation
independent jitter: sd = sqrt((T-1)/12)
random walk: scale (1/2) log T (turns)
persistent bias: 0
spread of the winding (turns)
0
2
4
6
8
10^1
10^2
10^3
length of the tie T (steps)